

CHAPTER 2

Basic Structure

CONTENTS

2.1 Sets

2.2 Set Operations

2.3 Functions.

2.1 Sets

2.1.1 Definitions and Notation

2.1.2 Cardinality

2.1.3 Power Set

2.1.4 Cartesian Products

2.1.1 Definitions and notation

Definition. A *set* is an unordered collection of elements.

Examples.

$\{1, 2, 3\}$ is the set containing “1” and “2” and “3.”

$\{1, 1, 2, 3, 3\} = \{1, 2, 3\}$ since repetition is irrelevant.

$\{1, 2, 3\} = \{3, 2, 1\}$ since sets are unordered.

$\{1, 2, 3, \dots\}$ is a way we denote an infinite set (in this case, the natural numbers).

$\emptyset = \{\}$ is the empty set, or the set containing no element.

Note: $\emptyset \neq \{\emptyset\}$

2.1.1 Definitions and notation

$x \in S$ means “ x is an element of set S .”

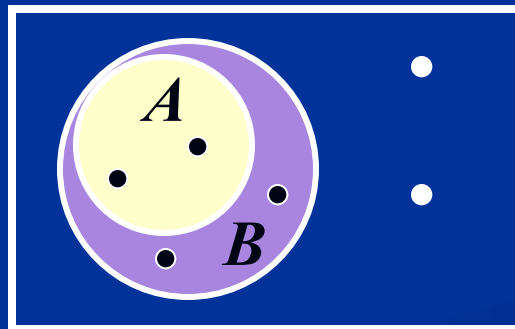
$x \notin S$ means “ x is not an element of set S .”

$A \subseteq B$ means “ A is a subset of B .”

or, “ B contains A .”

or, “every element of A is also in B .”

or, $\forall x ((x \in A) \rightarrow (x \in B))$.



Venn Diagram

2.1.1 Definitions and notation

$A \subseteq B$ means “ A is a subset of B .”

$A \supseteq B$ means “ A is a superset of B .”

$A = B$ if and only if A and B have exactly the same elements

iff, $A \subseteq B$ and $B \subseteq A$

iff, $A \subseteq B$ and $A \supseteq B$

iff, $\forall x ((x \in A) \leftrightarrow (x \in B))$.

So to show equality of sets A and B , show:

$$A \subseteq B$$

$$B \subseteq A$$

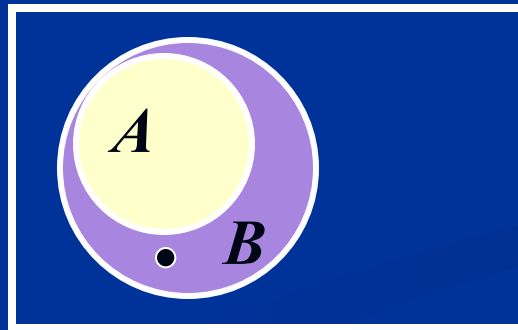
2.1.1 Definitions and notation

$A \subset B$ means “ A is a proper subset of B . ”

- $A \subseteq B$, and $A \neq B$.

- $\forall x ((x \in A) \rightarrow (x \in B))$

$\wedge \exists x ((x \in B) \wedge (x \notin A))$



2.1.1 Definitions and notation

Quick examples:

- $\{1,2,3\} \subseteq \{1,2,3,4,5\}$
- $\{1,2,3\} \subset \{1,2,3,4,5\}$

Is $\emptyset \subseteq \{1,2,3\}$?

Yes! $\forall x (x \in \emptyset) \rightarrow (x \in \{1,2,3\})$ holds,
because $(x \in \emptyset)$ is false.

Is $\emptyset \in \{1,2,3\}$? **No!**

Is $\emptyset \subseteq \{\emptyset, 1, 2, 3\}$? **Yes!**

Is $\emptyset \in \{\emptyset, 1, 2, 3\}$? **Yes!**

2.1.1 Definitions and notation

Quiz time:

Is $\{x\} \subseteq \{x\}$?

Yes

Is $\{x\} \in \{x, \{x\}\}$?

Yes

Is $\{x\} \subseteq \{x, \{x\}\}$?

Yes

Is $\{x\} \in \{x\}$?

No

Ways to define sets

- Explicitly: {John, Paul, George, Ringo}
- Implicitly: {1,2,3,...}, or {2,3,5,7,11,13,17,...}
- Set builder: { $x : x$ is prime }, { $x \mid x$ is odd }.

Ways to define sets

In general $\{ x : P(x) \}$, where $P(x)$ is some predicate

We read
“the set of all x such that $P(x)$ ”

Cardinality

If S is finite, then the *cardinality* of S , $|S|$, is the number of distinct elements in S .

If $S = \{1, 2, 3\}$

$$|S| = 3.$$

If $S = \{3, 3, 3, 3, 3\}$

$$|S| = 1.$$

If $S = \emptyset$

$$|S| = 0.$$

If $S = \{ \emptyset, \{ \emptyset \}, \{ \emptyset, \{ \emptyset \} \} \}$

$$|S| = 3.$$

If $S = \{0, 1, 2, 3, \dots\}$, $|S|$ is infinite. (more on this later)

Power sets

If S is a set, then the **power set** of S is

$$P(S) = 2^S = \{ x : x \subseteq S \}.$$

We say, “ $P(S)$ is the set of all subsets of S .”

If $S = \{a\}$ $2^S = \{\emptyset, \{a\}\}.$

If $S = \{a, b\}$ $2^S = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}.$

If $S = \emptyset$ $2^S = \{\emptyset\}.$

If $S = \{\emptyset, \{\emptyset\}\}$ $2^S = \{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\emptyset\}\}\}.$

Fact: if S is finite, $|2^S| = 2^{|S|}$. (if $|S| = n$, $|2^S| = 2^n$)

Cartesian Product

The **Cartesian Product** of two sets A and B is:

$$A \times B = \{ (a, b) : a \in A \wedge b \in B \}$$

If $A = \{\text{Charlie, Lucy, Linus}\}$, and
 $B = \{\text{Brown, VanPelt}\}$, then

We'll use these special
sets soon!

$$A \times B = \{(\text{Charlie, Brown}), (\text{Lucy, Brown}), (\text{Linus, Brown}),$$

 $(\text{Charlie, VanPelt}), (\text{Lucy, VanPelt}), (\text{Linus, VanPelt})\}$

Note. $(a, b) = (c, d)$ iff $a = c$, and $b = d$

$$A, B \text{ finite} \rightarrow |A \times B| = |A||B|$$

Cartesian Product

The **Cartesian Product** of n sets $A_1, A_2, \dots, A_n$ is:

$$\begin{aligned} A_1 \times A_2 \times \dots \times A_n &= \\ &= \{(a_1, a_2, \dots, a_n) : a_1 \in A_1, a_2 \in A_2, \dots, a_n \in A_n\} \end{aligned}$$

Note. $(a_1, a_2, \dots, a_n) = (b_1, b_2, \dots, b_n)$ iff
 $a_1 = b_1, a_2 = b_2, \dots, a_n = b_n$

We'll use these special
sets soon!

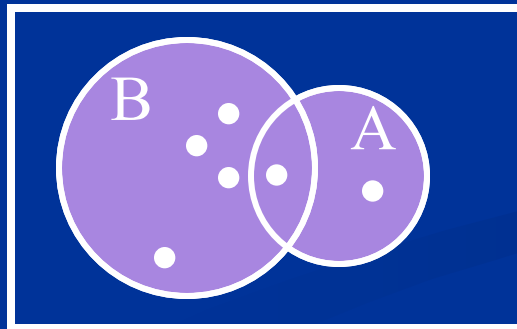
2.2.1 Set operation

The **union** of two sets A and B is:

$$A \cup B = \{x : x \in A \vee x \in B\}$$

If $A = \{\text{Charlie, Lucy, Linus}\}$, and
 $B = \{\text{Lucy, Desi}\}$, then

$$A \cup B = \{\text{Charlie, Lucy, Linus, Desi}\}$$



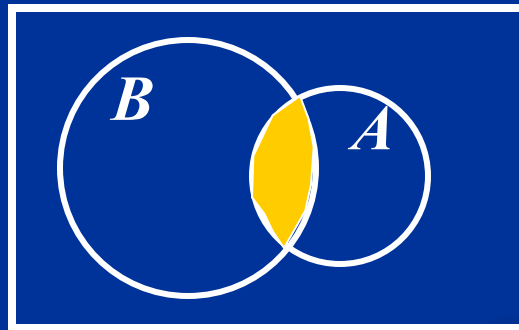
The *intersection* of two sets A and B is:

$$A \cap B = \{x : x \in A \wedge x \in B\}$$

If $A = \{\text{Charlie, Lucy, Linus}\}$, and

$B = \{\text{Lucy, Desi}\}$, then

$$A \cap B = \{\text{Lucy}\}$$



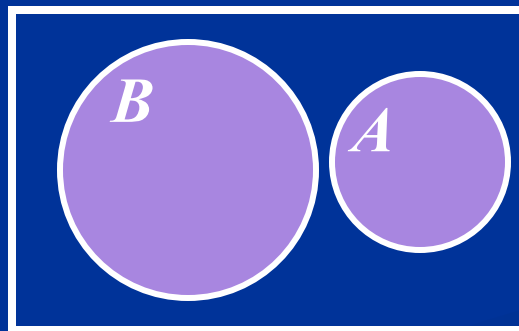
The *intersection* of two sets A and B is:

$$A \cap B = \{x : x \in A \wedge x \in B\}$$

If $A = \{x : x \text{ is a US president}\}$, and

$B = \{x : x \text{ is in this room}\}$, then

$$A \cap B = \{x : x \text{ is a US president in this room}\} = \emptyset$$



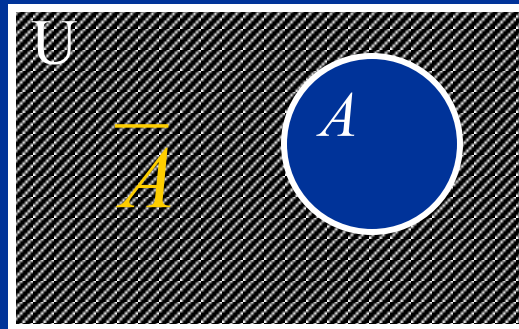
Sets whose
intersection is
empty are called
disjoint sets

The *complement* of a set A is:

$$\overline{A} = \{x : x \notin A\}$$

If $A = \{x : x \text{ is not shaded}\}$, then

$$\overline{A} = \{x : x \text{ is shaded}\}$$



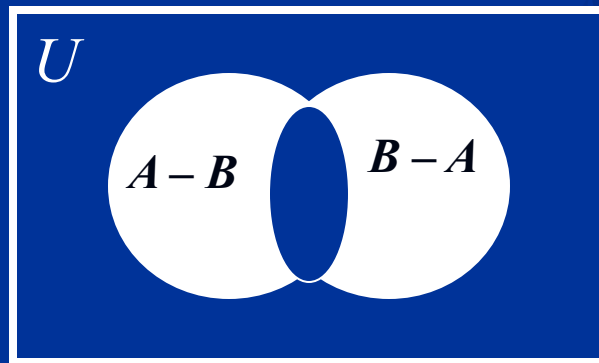
$$\begin{aligned} \overline{\emptyset} &= U \\ \text{and} \\ U &= \overline{\emptyset} \end{aligned}$$

The *symmetric difference*, $A \oplus B$, is:

$$A \oplus B = \{x : (x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin A)\}$$

$$= (A - B) \cup (B - A)$$

$$= \{x : x \in A \oplus x \in B\}$$



2.2.2 Set Identities

■ *Identity*

$$A \cap U = A$$

$$A \cup \emptyset = A$$

■ *Domination*

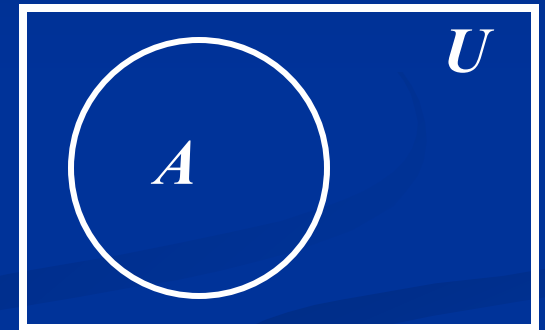
$$A \cup U = U$$

$$A \cap \emptyset = \emptyset$$

■ *Idempotent*

$$A \cup A = A$$

$$A \cap A = A$$



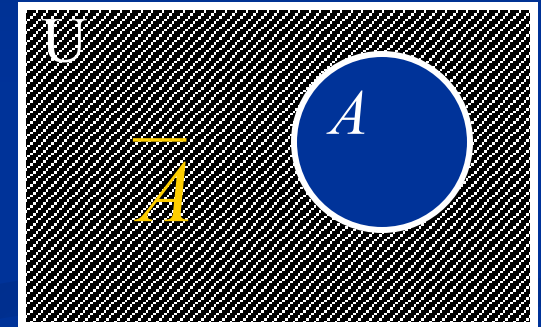
2.2.2 Set Identities

■ *Excluded Middle* $A \cup \bar{A} = U$

■ *Uniqueness*

$$A \cap \bar{A} = \emptyset$$

■ *Double complement* $\bar{\bar{A}} = A$



2.2.2 Set Identities

■ *Commutativity*

$$A \cup B = B \cup A$$

$$A \cap B = B \cap A$$

■ *Associativity*

$$(A \cup B) \cup C = A \cup (B \cup C)$$

$$(A \cap B) \cap C = A \cap (B \cap C)$$

■ *Distributivity*

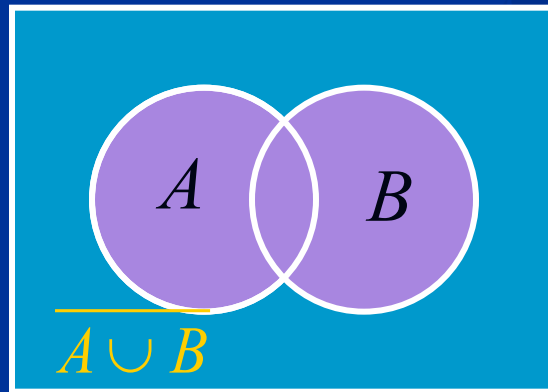
$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

2.2.2 Set Identities

■ *DeMorgan's I* $\overline{A \cup B} = \bar{A} \cap \bar{B}$

■ *DeMorgan's II* $\overline{A \cap B} = \bar{A} \cup \bar{B}$



4 Ways to prove identities

- Show that $A \subseteq B$ and that $A \supseteq B$.

New &
important

- Use a membership table.

Like truth tables

- Use previously proven identities.

Like $\equiv$

- Use logical equivalences to prove equivalent set definitions.

Not hard, a little tedious

4 Ways to prove identities

Prove that $\overline{A \cup B} = \overline{A} \cap \overline{B}$

$$\begin{aligned}x \in \overline{A \cup B} &\equiv x \notin A \cup B \equiv \neg(x \in A \vee x \in B) \\&\equiv (\neg x \in A) \wedge (\neg x \in B) \equiv x \notin A \wedge x \notin B \\&\equiv x \in \overline{A} \wedge x \in \overline{B} \equiv x \in \overline{A} \cap \overline{B}\end{aligned}$$

Not a particularly
interesting example,
sorry.

4 Ways to prove identities

Prove that $\overline{A \cup B} = \bar{A} \cap \bar{B}$

using logically equivalent set definitions

$$\begin{aligned}\overline{(A \cup B)} &= \{x : \neg(x \in A \vee x \in B)\} \\ &= \{x : \neg(x \in A) \wedge \neg(x \in B)\} \\ &= \{x : (x \in \bar{A}) \wedge (x \in \bar{B})\} \\ &= \bar{A} \cap \bar{B}\end{aligned}$$

4 Ways to prove identities

Prove that $\overline{A \cup (B \cap C)} = (\overline{C} \cup \overline{B}) \cap \overline{A}$

using known identities

$$\begin{aligned}\overline{A \cup (B \cap C)} &= \overline{A} \cap \overline{(B \cap C)} \\ &= \overline{A} \cap (\overline{B} \cup \overline{C}) \\ &= (\overline{B} \cup \overline{C}) \cap \overline{A} \\ &= (\overline{C} \cup \overline{B}) \cap \overline{A}\end{aligned}$$

35. Show that $A \oplus B = (A \cup B) - (A \cap B)$.

36. Show that $A \oplus B = (A - B) \cup (B - A)$.

37. Show that if A is a subset of a universal set U , then

a) $A \oplus A = \emptyset$.

b) $A \oplus \emptyset = A$.

c) $A \oplus U = \overline{A}$.

d) $A \oplus \overline{A} = U$.

2.2.3 Generalized Set Operations

Generalized Union

$$\bigcup_{i=1}^n A_i = A_1 \cup A_2 \cup \dots \cup A_n$$
$$= \{x : x \in A_1 \vee x \in A_2 \vee \dots \vee x \in A_n\}$$

Ex. Let $U = \mathbb{N}$, and define:

$$A_i = \{i, i+1, i+2, \dots\}$$

Then

$$\bigcup_{i=1}^n A_i = \bigcup_{i=1}^n \{i, i+1, i+2, \dots\} = \{1, 2, 3, \dots\}$$

Generalized Intersection

$$\bigcap_{i=1}^n A_i = A_1 \cap A_2 \cap \dots \cap A_n$$

$$= \{x : x \in A_1 \wedge x \in A_2 \wedge \dots \wedge x \in A_n\}$$

Ex. Let $U = \mathbf{N}$, and define:

$$A_i = \{i, i+1, i+2, \dots\}$$

Then

$$\bigcap_{i=1}^n A_i = \{n, n+1, n+2, \dots\}$$

47. Let $A_i = \{1, 2, 3, \dots, i\}$ for $i = 1, 2, 3, \dots$. Find

a) $\bigcup_{i=1}^n A_i.$

b) $\bigcap_{i=1}^n A_i.$

48. Let $A_i = \{\dots, -2, -1, 0, 1, \dots, i\}$. Find

a) $\bigcup_{i=1}^n A_i.$

b) $\bigcap_{i=1}^n A_i.$

51. Find $\bigcup_{i=1}^{\infty} A_i$ and $\bigcap_{i=1}^{\infty} A_i$ if for every positive integer i ,

a) $A_i = \{-i, -i + 1, \dots, -1, 0, 1, \dots, i - 1, i\}.$

b) $A_i = \{-i, i\}.$

c) $A_i = [-i, i]$, that is, the set of real numbers x with $-i \leq x \leq i.$

d) $A_i = [i, \infty)$, that is, the set of real numbers x with $x \geq i.$

2.2.4 Computer Representation of Sets

Let $U = \{x_1, x_2, \dots, x_n\}$, and choose an arbitrary order of the elements of U , say

$$x_1, x_2, \dots, x_n$$

Let $A \subseteq U$. Then the *bit string representation* of A is the bit string of length $n : a_1 a_2 \dots a_n$ such that $a_i = 1$ if $x_i \in A$, and 0 otherwise.

Ex. If $U = \{x_1, x_2, \dots, x_6\}$, and $A = \{x_1, x_3, x_5, x_6\}$,

then the bit string representation of A is

(101011)

Sets as bit strings

Ex. If $U = \{x_1, x_2, \dots, x_6\}$, $A = \{x_1, x_3, x_5, x_6\}$,
and $B = \{x_2, x_3, x_6\}$.

Then we have a quick way of finding the bit string
corresponding to $A \cup B$ and $A \cap B$.

Bit-wise OR

Bit-wise AND

A	1	0	1	0	1	1
B	0	1	1	0	0	1
$A \cup B$	1	1	1	0	1	1
$A \cap B$	0	0	1	0	0	1

52. Suppose that the universal set is $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Express each of these sets with bit strings where the i th bit in the string is 1 if i is in the set and 0 otherwise.

a) $\{3, 4, 5\}$

b) $\{1, 3, 6, 10\}$

c) $\{2, 3, 4, 7, 8, 9\}$

53. Using the same universal set as in the last problem, find the set specified by each of these bit strings.

a) 11 1100 1111

b) 01 0111 1000

c) 10 0000 0001

2.3 Functions

2.3.1 Introduction

2.3.2 One-to-One and Onto Functions.

2.3.3 Inverse Functions and
Composition of Functions

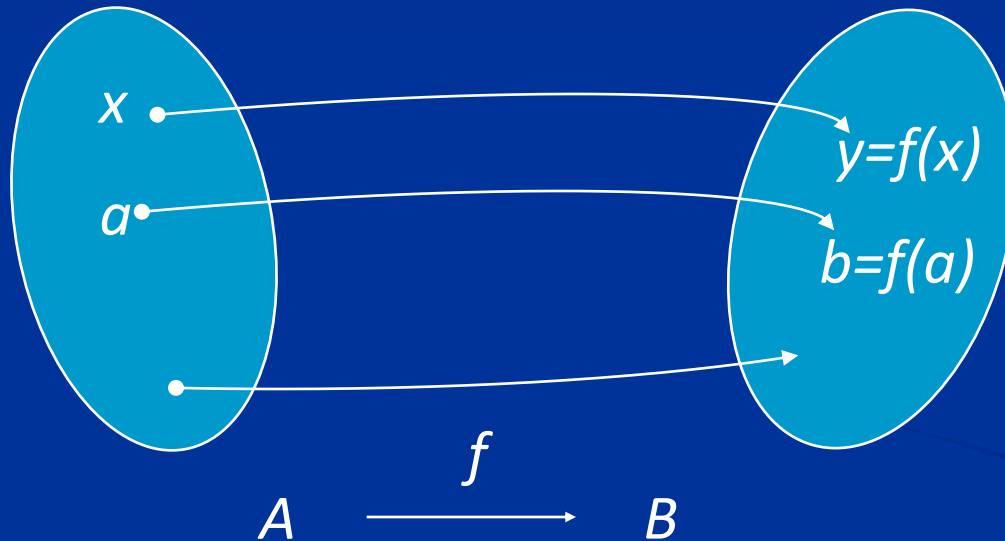
2.3.4 The Graphs of Functions

2.3.5 Asymptotic Notation

2.3.6 Hash function

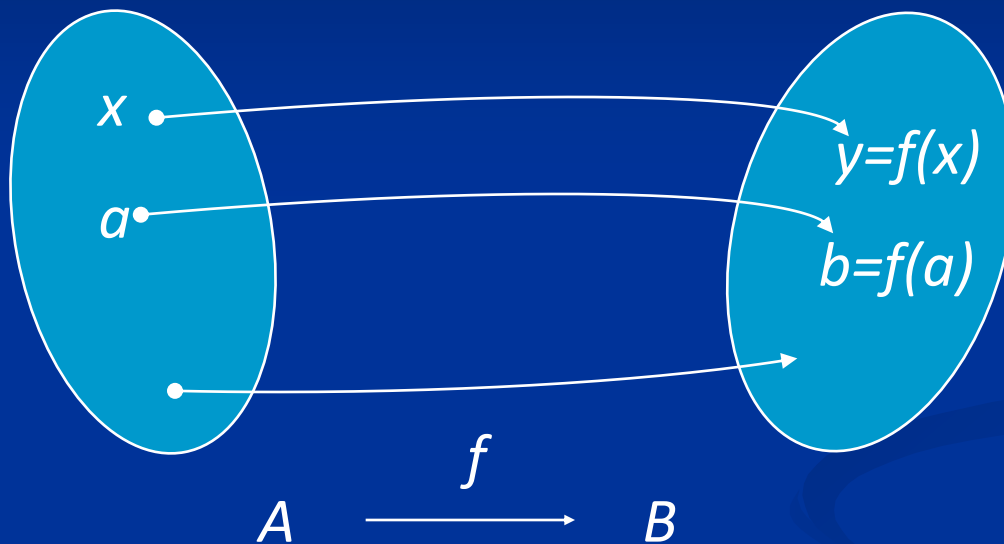
2.3.1 Introduction

Definition. A *function* f is a rule that assigns to each element x in a set A exactly one element $y=f(x)$ in a set B



- A is the *domain*, B is the *codomain* of f .

2.3.1 Introduction



- $b = f(a)$ is the *image* of a and a is the *preimage* of b .
- The *range* of f is the set $\{f(a), a \in A\}$

Algebra of functions: let f and g be functions with domains A and B . Then the functions $f+g$, $f-g$, fg and f/g are defined as follows:

$$(f + g)(x) = f(x) + g(x) \quad \text{domain} = A \cap B$$

$$(f - g)(x) = f(x) - g(x) \quad \text{domain} = A \cap B$$

$$(fg)(x) = f(x)g(x) \quad \text{domain} = A \cap B$$

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} \quad \text{domain} = \{x \in A \cap B \mid g(x) \neq 0\}$$

Example: let $f(x) = x^2$, $g(x) = x - x^2$ be functions from $\mathbb{R}$ to $\mathbb{R}$, find $f + g$ and fg .

Solution: We have

$$(f + g)(x) = x^2 + (x - x^2) = x$$

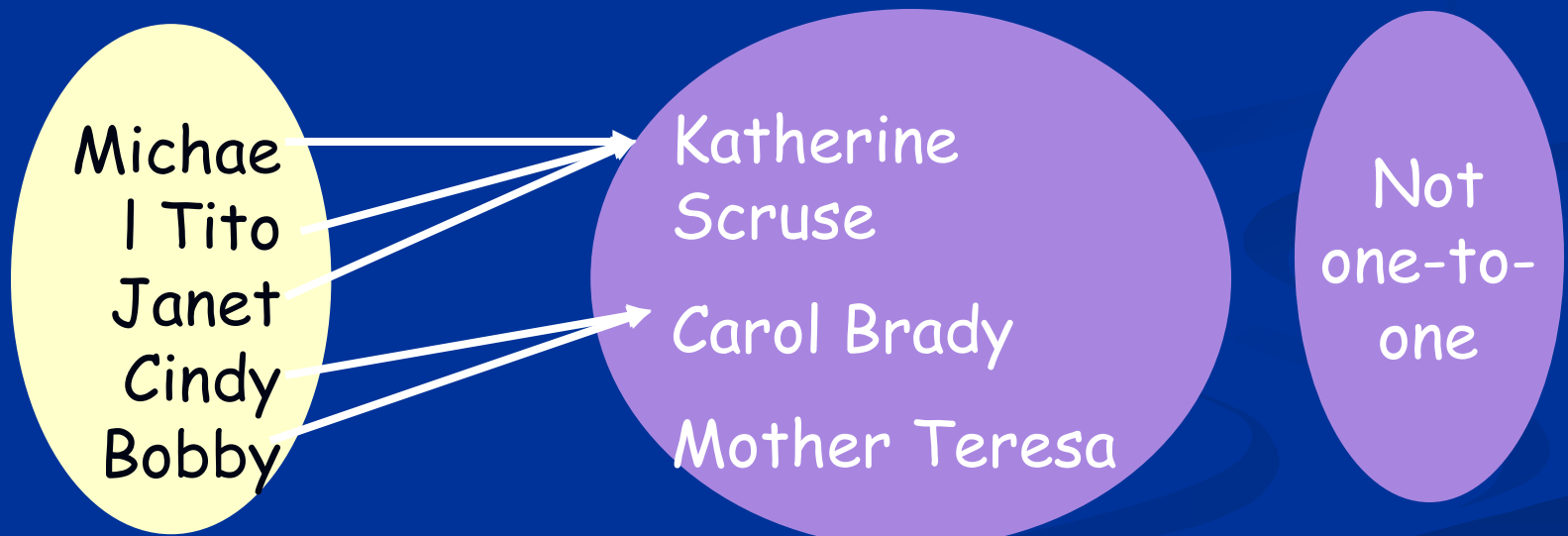
And

$$(fg)(x) = x^2 (x - x^2) = x^3 - x^4$$

2.3.2 One-to-One and Onto Functions

Definition. A function $f: A \rightarrow B$ is *one-to-one* (*injective, an injection*) if

$$\forall x, y (f(x) = f(y) \rightarrow x = y)$$



Every $b \in B$ has at most 1 preimage.

Remark. A function $f: A \rightarrow B$ is *one-to-one* iff
$$\forall x, y (x \neq y \rightarrow f(x) \neq f(y))$$

Recall that

❖ A function f is **strictly increasing** on an interval $I \subseteq \mathbf{R}$ if

$$\forall x, y (x < y \rightarrow f(x) < f(y))$$

❖ f is strictly **decreasing** on I if

$$\forall x, y (x < y \rightarrow f(x) > f(y))$$

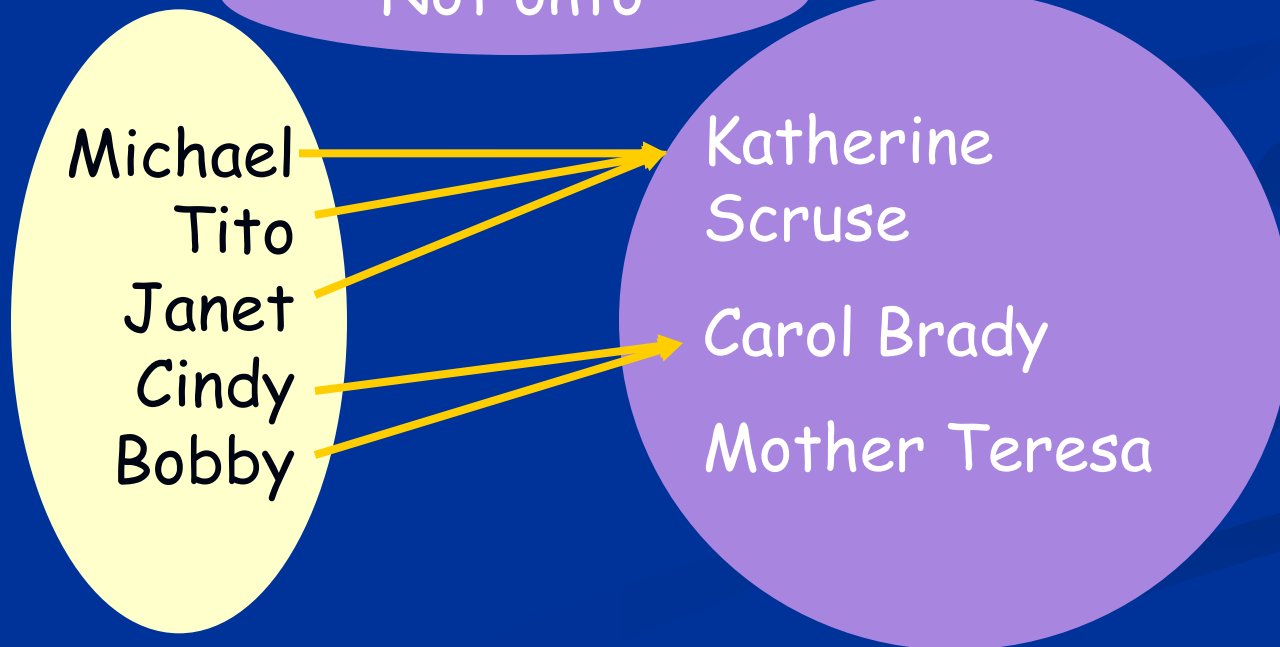
❖ It is clear that a strictly increasing or strictly decreasing function is one-to-one.

Onto Functions

Definition. A function $f: A \rightarrow B$ is *onto* (*surjective*, a *surjection*) if $\forall b \in B, \exists a \in A f(a) = b$

Every $b \in B$ has at least 1 preimage.

Not onto



Onto Functions

Example. Is the function $f(x) = x^2$ from $\mathbf{Z}$ to $\mathbf{Z}$ onto?

Solution: The function f is not onto since there is no x in $\mathbf{Z}$ such that $x^2 = -1$

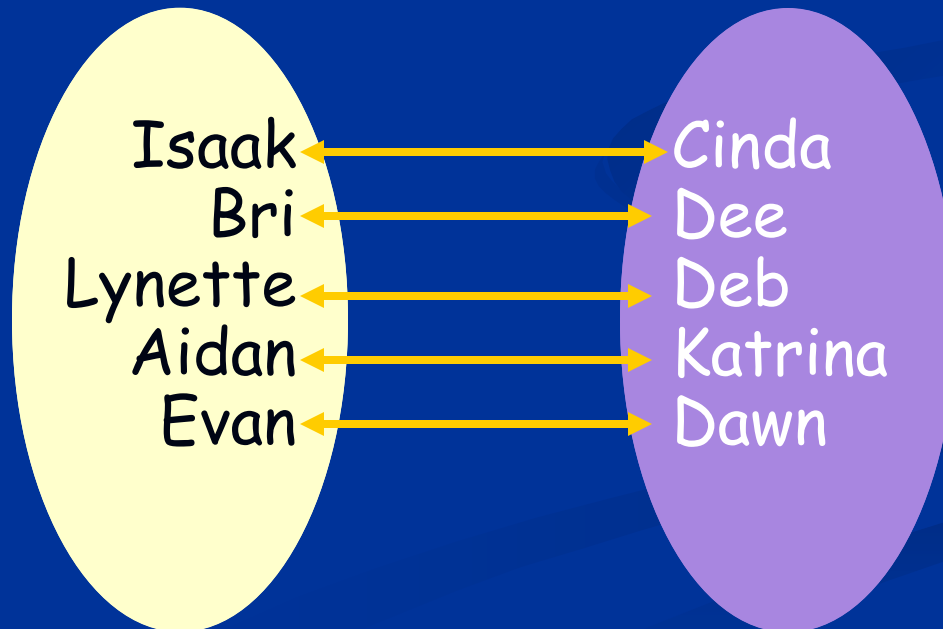
Example. Is the function $f(x) = x + 1$ from $\mathbf{Z}$ to $\mathbf{Z}$ onto?

Solution: The function f is onto since for every y in $\mathbf{Z}$, there is an element x in $\mathbf{Z}$ such that $x + 1 = y$ (by taking $x = y - 1$)

Bijection

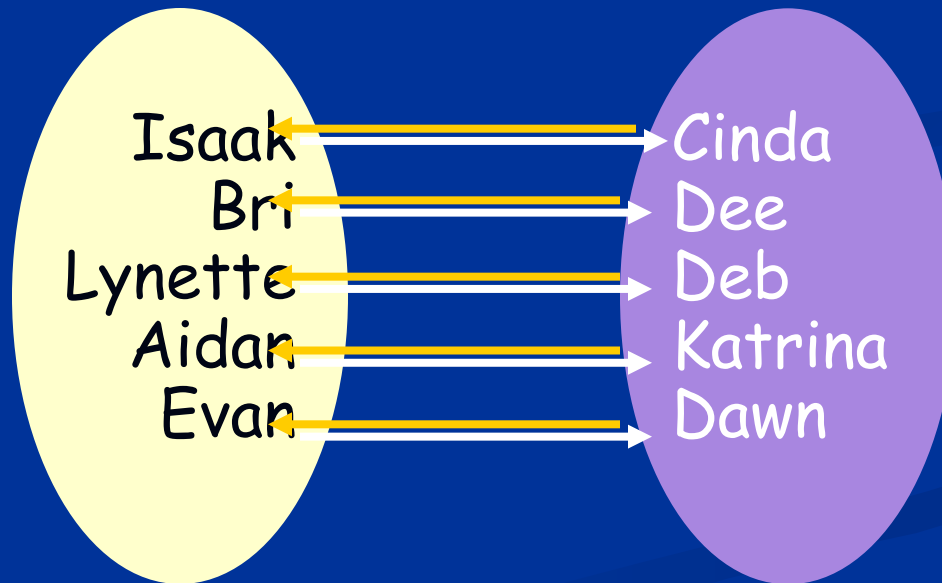
Definition. A function $f: A \rightarrow B$ is bijective if it is one-to-one and onto. We also say that f is a **bijection**

Every $b \in B$ has exactly 1 preimage.



2.3.3 Inverse Functions and Compositions of Functions

Definition. Let $f: A \rightarrow B$ be a bijection. Then the inverse function of f , denoted by f^{-1} is the function that assigns each element b in B the unique element a in A such that $f(a) = b$. Thus $f^{-1}(b) = a$.



$$f^{-1}(\text{Cinda}) = \text{Isaak}, f^{-1}(\text{Dee}) = \text{Bri}, \dots, f^{-1}(\text{Dawn}) = \text{Evan}$$

Example. Is the function $f(x) = x^2$ from $\mathbf{Z}$ to $\mathbf{Z}$ invertible? (i.e. the inverse function exists)

Solution: The function f is not onto. Therefore it is not a bijection, and hence not invertible

Example. Is the function $f(x) = x + 1$ from $\mathbf{Z}$ to $\mathbf{Z}$ invertible?

Solution: The function f is a bijection so it is invertible.

Example. Is the function $f(x) = x + 1$ from $\mathbf{Z}$ to $\mathbf{Z}$ invertible? What is its inverse?

Solution: The function f is a bijection so it is invertible.

To find the inverse, let y be any element in $\mathbf{Z}$, we find the element x in $\mathbf{Z}$ such that $y = f(x) = x + 1$.

Solving this equation we obtain $x = y - 1$.

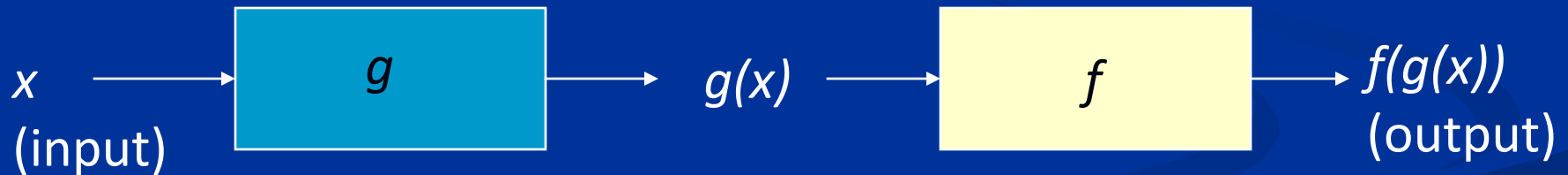
Hence $f^{-1}(y) = y - 1$.

We also write $f^{-1}(x) = x - 1$.

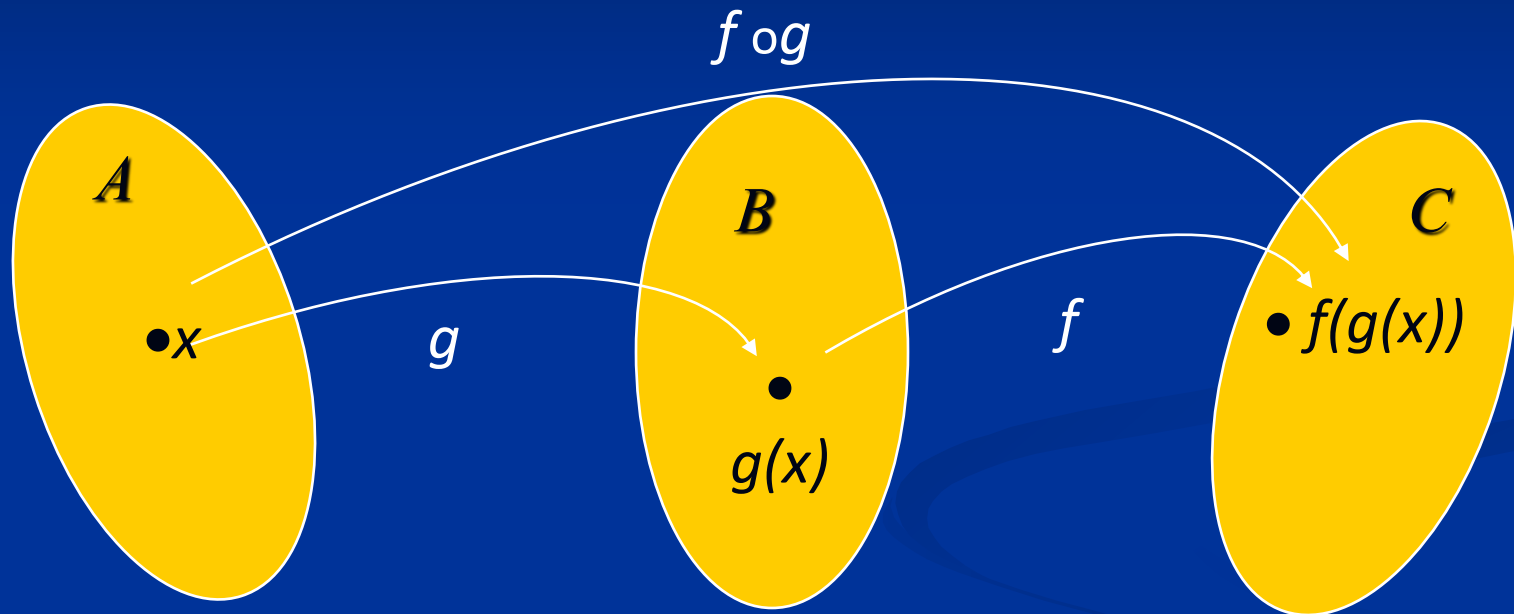
Definition. The composition of a function $g: A \rightarrow B$ and a function $f: B \rightarrow C$ is the function $f \circ g: A \rightarrow C$ defined by

$$f \circ g(x) = f(g(x))$$

Note. The domain of $f \circ g$ is also the domain of g , and the codomain of $f \circ g$ is also the codomain of f .



Arrow diagram for $f \circ g$



Example. let $f(x) = x^2$ and $g(x) = x - 3$ are functions from $\mathbf{R}$ to $\mathbf{R}$.

Find the compositions $f \circ g$ and $g \circ f$

Solution.

$$(f \circ g)(x) = f(g(x)) = f(x - 3) = (x - 3)^2$$

$$(g \circ f)(x) = g(f(x)) = g(x^2) = x^2 - 3$$

This shows that in general: $f \circ g \neq g \circ f$

12. Determine whether each of these functions from $\mathbf{Z}$ to $\mathbf{Z}$ is one-to-one.

a) $f(n) = n - 1$

b) $f(n) = n^2 + 1$

c) $f(n) = n^3$

d) $f(n) = \lceil n/2 \rceil$

13. Which functions in Exercise 12 are onto?

14. Determine whether $f: \mathbf{Z} \times \mathbf{Z} \rightarrow \mathbf{Z}$ is onto if

a) $f(m, n) = 2m - n.$

b) $f(m, n) = m^2 - n^2.$

c) $f(m, n) = m + n + 1.$

d) $f(m, n) = |m| - |n|.$

e) $f(m, n) = m^2 - 4.$

15. Determine whether the function $f: \mathbf{Z} \times \mathbf{Z} \rightarrow \mathbf{Z}$ is onto if

a) $f(m, n) = m + n.$

b) $f(m, n) = m^2 + n^2.$

c) $f(m, n) = m.$

d) $f(m, n) = |n|.$

e) $f(m, n) = m - n.$

22. Determine whether each of these functions is a bijection from $\mathbf{R}$ to $\mathbf{R}$.

a) $f(x) = -3x + 4$

b) $f(x) = -3x^2 + 7$

c) $f(x) = (x + 1)/(x + 2)$

d) $f(x) = x^5 + 1$

23. Determine whether each of these functions is a bijection from $\mathbf{R}$ to $\mathbf{R}$.

a) $f(x) = 2x + 1$

b) $f(x) = x^2 + 1$

c) $f(x) = x^3$

d) $f(x) = (x^2 + 1)/(x^2 + 2)$

2.3.5 Some Important Functions

Ceiling.

$f(x) = \lceil x \rceil$ the least integer y so that $x \leq y$.

Ex: $\lceil 1.2 \rceil = 2$; $\lceil -1.2 \rceil = -1$; $\lceil 1 \rceil = 1$

Floor.

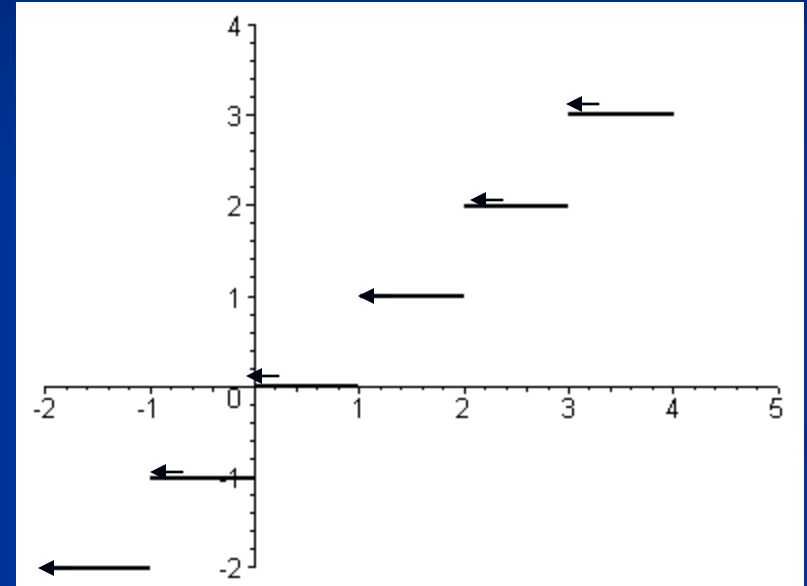
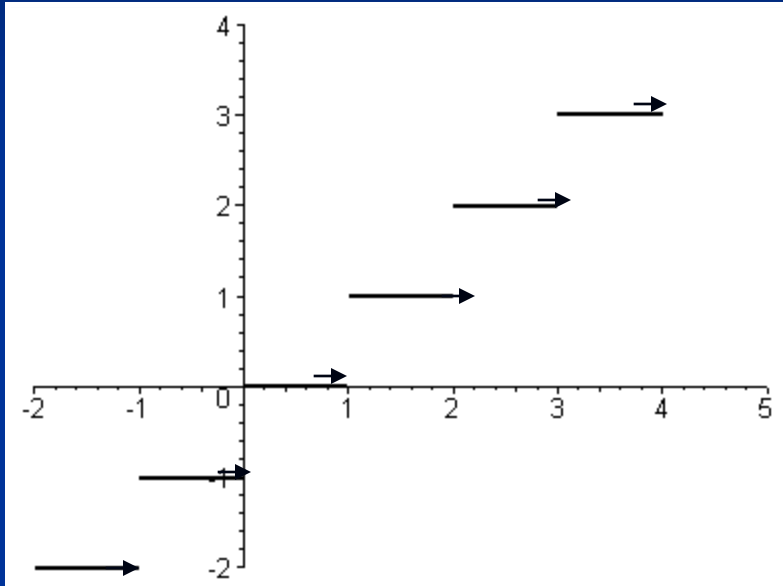
$f(x) = \lfloor x \rfloor$ the greatest integer y so that $y \leq x$.

Ex: $\lfloor 1.8 \rfloor = 1$; $\lfloor -1.8 \rfloor = -2$; $\lfloor -5 \rfloor = -5$

Quiz: what is $\lceil -1.2 + \lfloor 1.1 \rfloor \rceil$?

0

The graph of the floor function



The graph of the ceiling function

18. Show that if n is an integer, then $n = \lceil n/2 \rceil + \lfloor n/2 \rfloor$.
19. For which real numbers x and y is it true that $\lfloor x + y \rfloor = \lfloor x \rfloor + \lfloor y \rfloor$?
20. For which real numbers x and y is it true that $\lceil x + y \rceil = \lceil x \rceil + \lceil y \rceil$?
21. For which real numbers x and y is it true that $\lceil x + y \rceil = \lceil x \rceil + \lfloor y \rfloor$?
22. Prove that $\lfloor n/2 \rfloor \lceil n/2 \rceil = \lfloor n^2/4 \rfloor$ for all integers n .
23. Prove that if m is an integer, then $\lfloor x \rfloor + \lfloor m - x \rfloor = m - 1$, unless x is an integer, in which case, it equals m .
24. Prove that if x is a real number, then $\lfloor \lfloor x/2 \rfloor / 2 \rfloor = \lfloor x/4 \rfloor$.
25. Prove that if n is an odd integer, then $\lceil n^2/4 \rceil = (n^2 + 3)/4$.

43. Let $g(x) = \lfloor x \rfloor$. Find

a) $g^{-1}(\{0\})$.

b) $g^{-1}(\{-1, 0, 1\})$.

c) $g^{-1}(\{x \mid 0 < x < 1\})$.

44. Let f be a function from A to B . Let S and T be subsets of B . Show that

a) $f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T)$.

b) $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$.

Big-O, Big-Omega, Big-Theta notation

Let f and g be functions from the set of integers or the set of real numbers to the set of real numbers. We say that $f(x)$ is $O(g(x))$ if there are constants C and k such that

$$|f(x)| \leq C|g(x)|$$

whenever $x > k$. [This is read as “ $f(x)$ is big-oh of $g(x)$.”]

Let $f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$, where $a_0, a_1, \dots, a_{n-1}, a_n$ are real numbers. Then $f(x)$ is $O(x^n)$.

Proof: Using the triangle inequality (see Exercise 7 in Section 1.8), if $x > 1$ we have

$$\begin{aligned} |f(x)| &= |a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0| \\ &\leq |a_n| x^n + |a_{n-1}| x^{n-1} + \cdots + |a_1| x + |a_0| \\ &= x^n (|a_n| + |a_{n-1}|/x + \cdots + |a_1|/x^{n-1} + |a_0|/x^n) \\ &\leq x^n (|a_n| + |a_{n-1}| + \cdots + |a_1| + |a_0|). \end{aligned}$$

This shows that

$$|f(x)| \leq C x^n,$$

where $C = |a_n| + |a_{n-1}| + \cdots + |a_0|$ whenever $x > 1$. Hence, the witnesses $C = |a_n| + |a_{n-1}| + \cdots + |a_0|$ and $k = 1$ show that $f(x)$ is $O(x^n)$. 

Let $f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$, where $a_0, a_1, \dots, a_{n-1}, a_n$ are real numbers. Then $f(x)$ is $O(x^n)$.

Suppose that $f_1(x)$ is $O(g_1(x))$ and that $f_2(x)$ is $O(g_2(x))$. Then $(f_1 + f_2)(x)$ is $O(\max(|g_1(x)|, |g_2(x)|))$.

Suppose that $f_1(x)$ and $f_2(x)$ are both $O(g(x))$. Then $(f_1 + f_2)(x)$ is $O(g(x))$.

Suppose that $f_1(x)$ is $O(g_1(x))$ and $f_2(x)$ is $O(g_2(x))$. Then $(f_1 f_2)(x)$ is $O(g_1(x)g_2(x))$.

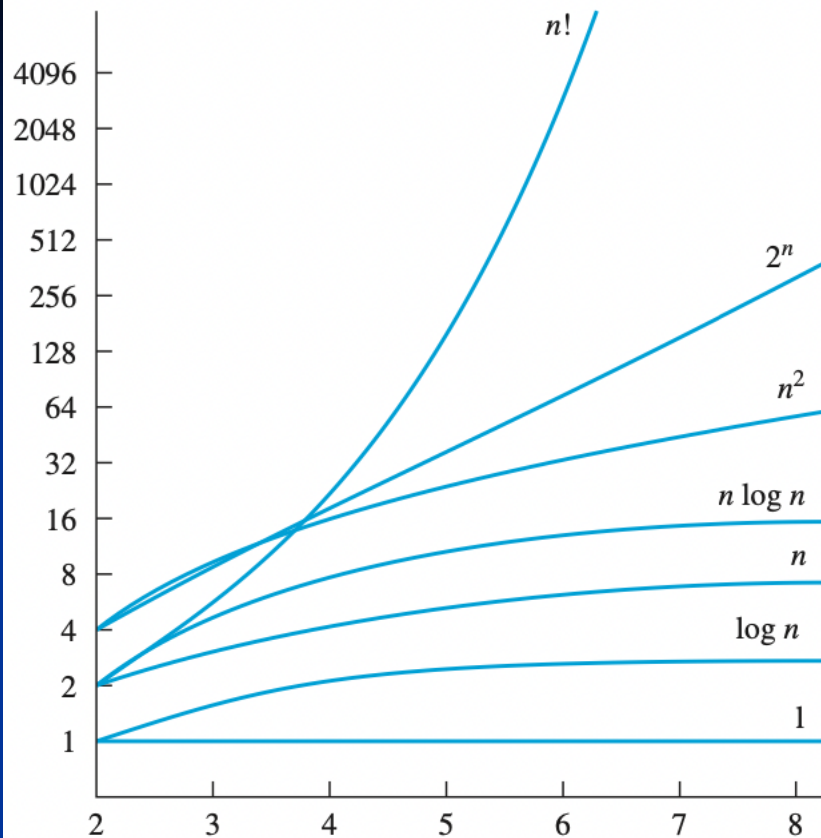


FIGURE 3 A Display of the Growth of Functions Commonly Used in Big- O Estimates.

The functions used in these estimates often include the following:

$1, \log n, n, n \log n, n^2, 2^n, n!$

EXAMPLE

Give a big- O estimate for $f(x) = (x + 1) \log(x^2 + 1) + 3x^2$.

Solution: First, a big- O estimate for $(x + 1) \log(x^2 + 1)$ will be found. Note that $(x + 1)$ is $O(x)$. Furthermore, $x^2 + 1 \leq 2x^2$ when $x > 1$. Hence,

$$\log(x^2 + 1) \leq \log(2x^2) = \log 2 + \log x^2 = \log 2 + 2 \log x \leq 3 \log x,$$

if $x > 2$. This shows that $\log(x^2 + 1)$ is $O(\log x)$.

From Theorem 3 it follows that $(x + 1) \log(x^2 + 1)$ is $O(x \log x)$. Because $3x^2$ is $O(x^2)$, Theorem 2 tells us that $f(x)$ is $O(\max(x \log x, x^2))$. Because $x \log x \leq x^2$, for $x > 1$, it follows that $f(x)$ is $O(x^2)$. 

Let f and g be functions from the set of integers or the set of real numbers to the set of real numbers. We say that $f(x)$ is $\Omega(g(x))$ if there are positive constants C and k such that

$$|f(x)| \geq C|g(x)|$$

whenever $x > k$. [This is read as “ $f(x)$ is big-Omega of $g(x)$.”]

Let f and g be functions from the set of integers or the set of real numbers to the set of real numbers. We say that $f(x)$ is $\Theta(g(x))$ if $f(x)$ is $O(g(x))$ and $f(x)$ is $\Omega(g(x))$. When $f(x)$ is $\Theta(g(x))$ we say that f is big-Theta of $g(x)$, that $f(x)$ is of *order* $g(x)$, and that $f(x)$ and $g(x)$ are of the *same order*.

Show that $3x^2 + 8x \log x$ is $\Theta(x^2)$.

Solution: Because $0 \leq 8x \log x \leq 8x^2$, it follows that $3x^2 + 8x \log x \leq 11x^2$ for $x > 1$. Consequently, $3x^2 + 8x \log x$ is $O(x^2)$. Clearly, x^2 is $O(3x^2 + 8x \log x)$. Consequently, $3x^2 + 8x \log x$ is $\Theta(x^2)$.

Let $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$, where $a_0, a_1, \dots, a_n$ are real numbers with $a_n \neq 0$. Then $f(x)$ is of order x^n .

The polynomials $3x^8 + 10x^7 + 221x^2 + 1444$, $x^{19} - 18x^4 - 10,112$, and $-x^{99} + 40,001x^{98} + 100,003x$ are of orders x^8 , x^{19} , and x^{99} , respectively.